

Labsheet- 10

End-to-End Data Analytics Project: Business Intelligence and Predictive Analytics using a Real-World Dataset

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 1: Load Dataset
df = pd.read_csv("train.csv")

print("Dataset Loaded Successfully!")
print("Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

# Step 2: Select Numeric Columns
num = df.select_dtypes(include=np.number)

print("\nNumeric Columns:")
print(num.columns.tolist())

# Step 3: Remove Missing Values
num = num.fillna(num.median())

# Step 4: Target Column
target = num.columns[-1]

X = num.drop(target, axis=1)
y = num[target]

print("\nTarget Column:", target)

# Step 5: Statistics
print("\nStatistical Summary:")
print(num.describe())

# Step 6: Correlation Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(num.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

# Step 7: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 8: Train Model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
# Step 9: Prediction
y_pred = model.predict(X_test)

# Step 10: Evaluation
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("\nModel Performance:")
print("MAE:", round(mae, 2))
print("RMSE:", round(rmse, 2))
print("R2 Score:", round(r2, 3))

# Step 11: Actual vs Predicted
plt.figure(figsize=(7, 5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual")
plt.ylabel("Predicted")
plt.title("Actual vs Predicted")
plt.show()

# Step 12: Conclusion
print("\nConclusion:")
print("End-to-end data analytics and prediction completed successfully.")
```

Dataset Loaded Successfully!

Shape: (4600, 18)

Columns:

['date', 'price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated', 'street', 'city', 'statezip', 'country']

Numeric Columns:

['price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated']

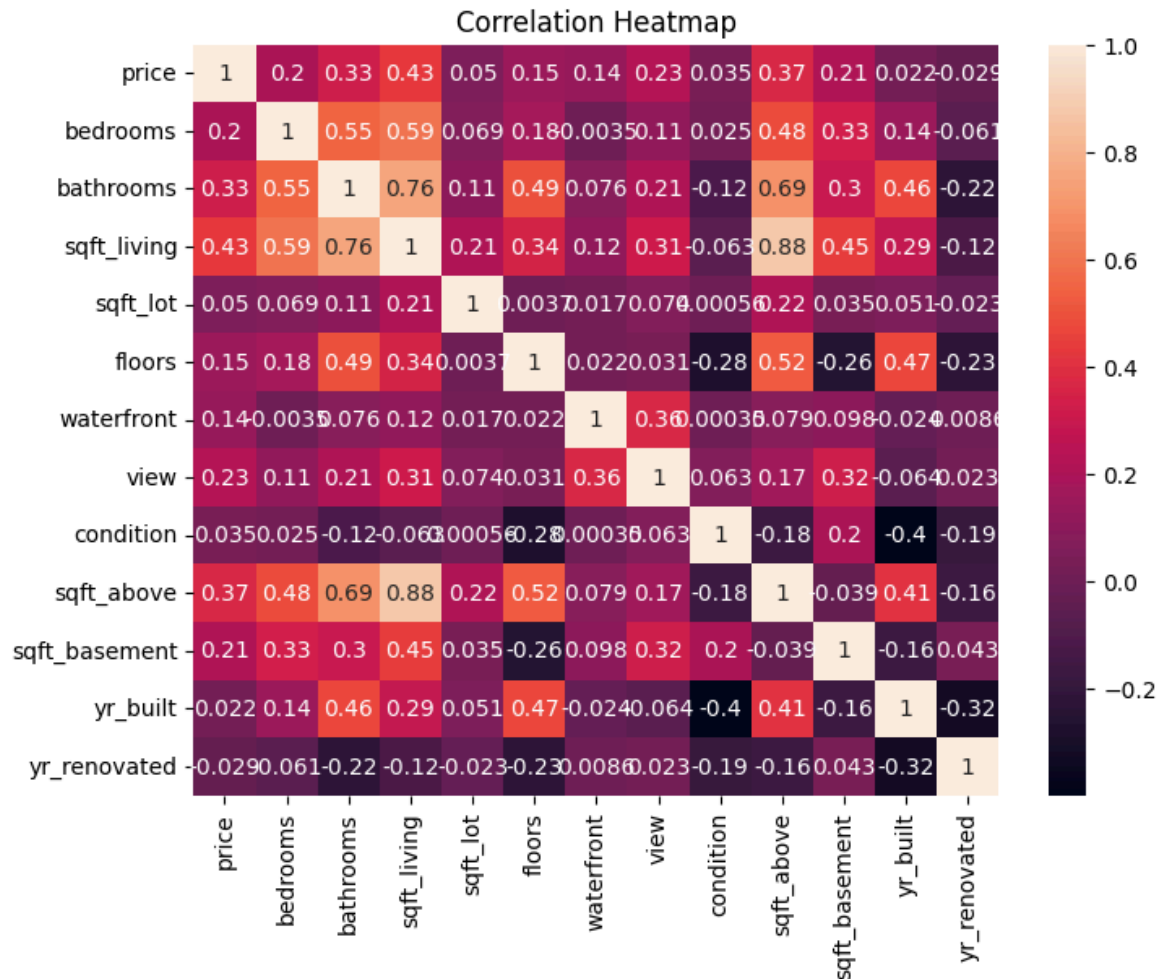
Target Column: yr_renovated

Statistical Summary:

	price	bedrooms	bathrooms	sqft_living	sqft_lot \
count	4.600000e+03	4600.000000	4600.000000	4600.000000	4.600000e+03
mean	5.519630e+05	3.400870	2.160815	2139.346957	1.485252e+04
std	5.638347e+05	0.908848	0.783781	963.206916	3.588444e+04
min	0.000000e+00	0.000000	0.000000	370.000000	6.380000e+02
25%	3.228750e+05	3.000000	1.750000	1460.000000	5.000750e+03
50%	4.609435e+05	3.000000	2.250000	1980.000000	7.683000e+03
75%	6.549625e+05	4.000000	2.500000	2620.000000	1.100125e+04
max	2.659000e+07	9.000000	8.000000	13540.000000	1.074218e+06

	floors	waterfront	view	condition	sqft_above \
count	4600.000000	4600.000000	4600.000000	4600.000000	4600.000000
mean	1.512065	0.007174	0.240652	3.451739	1827.265435
std	0.538288	0.084404	0.778405	0.677230	862.168977
min	1.000000	0.000000	0.000000	1.000000	370.000000
25%	1.000000	0.000000	0.000000	3.000000	1190.000000
50%	1.500000	0.000000	0.000000	3.000000	1590.000000
75%	2.000000	0.000000	0.000000	4.000000	2300.000000
max	3.500000	1.000000	4.000000	5.000000	9410.000000

	sqft_basement	yr_built	yr_renovated
count	4600.000000	4600.000000	4600.000000
mean	312.081522	1970.786304	808.608261
std	464.137228	29.731848	979.414536
min	0.000000	1900.000000	0.000000
25%	0.000000	1951.000000	0.000000
50%	0.000000	1976.000000	0.000000
75%	610.000000	1997.000000	1999.000000
max	4820.000000	2014.000000	2014.000000

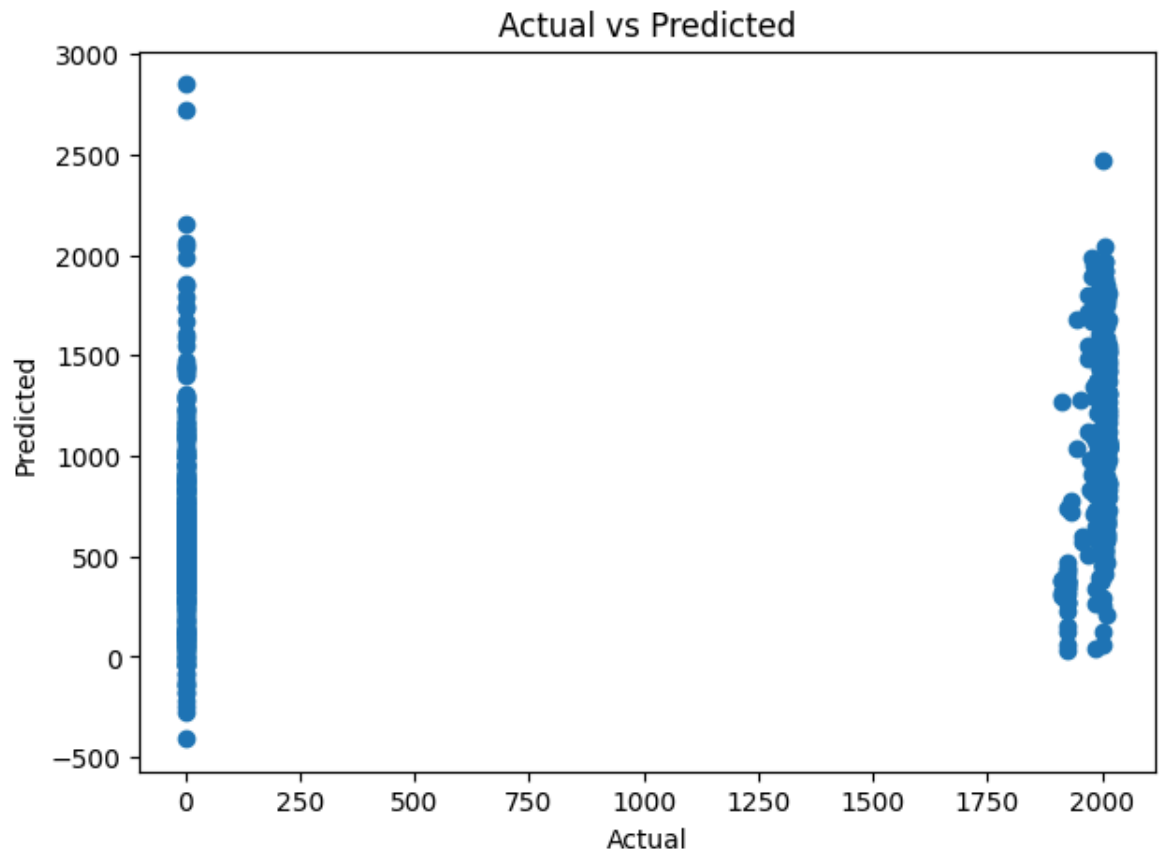


Model Performance:

MAE: 746.55

RMSE: 868.98

R2 Score: 0.216



Conclusion:

End-to-end data analytics and prediction completed successfully.

Questions and Answers**1. What are the major stages involved in an end-to-end data analytics project?**

An end-to-end data analytics project generally includes data collection, data preprocessing, exploratory data analysis, visualization, statistical analysis, machine learning, model evaluation, insight generation, and presentation of results.

2. Why is data preprocessing considered the foundation of successful analytics and machine learning?

Data preprocessing removes errors, handles missing values, removes duplicates, and converts data into a suitable format. Good-quality data improves the reliability and accuracy of analytics and machine learning models.

3. How does Exploratory Data Analysis (EDA) contribute to model development?

EDA helps understand the dataset, identify patterns, detect outliers, find relationships between variables, and select useful features. This helps in choosing suitable models and improving model performance.

4. What factors should be considered while selecting a machine learning algorithm for a business problem?

Important factors include the type of problem, dataset size, data type, number of features, model complexity, accuracy requirements, training time, interpretability, and available computing resources.

5. Why is model evaluation essential before deploying a predictive model?

Model evaluation measures how well a model performs on unseen data. Metrics such as Accuracy, Precision, Recall, F1-Score, MAE, RMSE, and R^2 help determine whether the model is reliable for practical use.

6. Explain how Business Intelligence tools complement machine learning in data analytics projects.

Business Intelligence tools such as Power BI and Tableau help present data and model results through interactive dashboards and reports. Machine learning provides predictions, while BI tools help users understand and monitor those predictions for decision-making.

7. What challenges are commonly encountered while working with real-world datasets?

Common challenges include missing values, duplicate records, incorrect data, outliers, inconsistent formats, categorical variables, large datasets, biased data, and difficulties in collecting reliable information.

8. How can analytical insights be converted into actionable business recommendations?

Insights should be connected to specific business objectives. Important trends and problems can be identified, their causes analyzed, and practical actions can then be suggested to improve sales, reduce costs, increase customer satisfaction, or improve efficiency.

9. Why is effective presentation and data storytelling important in analytics projects?

Data storytelling converts complex analysis into a clear and understandable message. It helps stakeholders understand important findings, make informed decisions, and take appropriate business actions.

10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution.

Future improvements can include feature engineering, hyperparameter tuning, ensemble models, cross-validation, advanced algorithms such as Random Forest and Gradient Boosting, larger datasets, automated dashboards, and real-time data analysis.